

Bohan Su

Wuhan University Software Engineering (Bachelor)

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Education

Wuhan University Bachelor Software Engineering **Sep 2023 – Jun 2027 (Expected)**

- **GPA:** 3.75 / 4.0
- **Core Courses:** Computer Vision, Frontier Technologies in AI, Computer Networks, Operating Systems, Data Structures and Algorithms, Discrete Mathematics, Probability Theory and Mathematical Statistics A, Software Engineering, Software Quality Assurance and Testing, Database Systems, OOP, AIoT Software Design and Security, Internet of Things Technology and Applications; Machine Learning for Computer Vision (Jens Rittscher, University of Oxford; 93.10 / 100)
- **Research Interests:** Large-model reasoning and machine learning, with a focus on applying structured representation and reasoning enhancement to computer vision and scientific problem solving

Research & Ongoing Work

Reliability-Aware 3D Geometric Injection for Universal Person Re-identification (First Author) Oct 2025–Mar 2026
ECCV 2026 Accept

- Targeting complex Universal ReID scenarios involving occlusion, clothing change, and cross-modality, completed the paper accepted by ECCV 2026 (ID: 10537) as first author, responsible for problem formulation, method design, experimental protocol, and writing of the main paper and supplementary material.
- Proposed the UniGeo framework, decoupling SMPL body parameters into global body-shape information and joint topology, and injecting monocular 3D geometric priors as appearance-independent cues into the ReID representation learning process.
- Designed a Reliability-Aware Gate that dynamically fuses 2D visual and 3D structural representations at the feature level, reducing the interference of low-confidence geometry estimates on identification features while maintaining lightweight adaptation to existing ReID backbones.
- On 9 ReID benchmarks including Market-1501 and MSMT17, conducted comparison, ablation, and image-perturbation stability experiments under a unified protocol, analyzing the benefit boundary of 3D geometric injection under occlusion, clothing-change, and cross-modality scenarios.

PEARL: Verifiable Curation and Small-Model Learning for Scientific Reasoning Graph Extraction (First Author) Mar 2026–Present
WAICA Wksp Accept
Target: AAAI Submission

- Positioned PEARL as an end-to-end framework for scientific reasoning graph extraction; the preliminary version accepted by WAICA Workshop decouples model-generated candidates from code-side checks for structure, compilability, and source alignment.
- Across five model archives and 350 samples, raised strict-gate passes from 0/350 to 300/350, average CG from 0.827 to 0.896, and REA from 0.339 to 0.906 through fresh-source evaluation, judge feedback, and provenance auditing.
- For the full AAAI-targeted version, using PEARL-curated graphs to supervise LoRA training of Qwen3.5-4B-class models through a nodes-to-reasoning-units-to-graph_spec-to-DOT curriculum.

Appearance Variations in Person Re-Identification: A Survey (First Author) Mar 2026–Present
Target: TPAMI

- Reconstructed the Person Re-ID survey framework around appearance variation, establishing a trichotomy of short-term perturbations, long-term appearance changes, and composite variations, and subdividing it into sub-problems such as structural geometry, spectral environment, information completeness, clothing state, appearance drift, and deployment evolution.
- Proposed an "appearance robustness vs. appearance generalization" methodological framework, reorganizing directions such as pose/viewpoint/resolution, RGB-IR, occlusion, clothing change, continual learning, vision-language models, and causal intervention, and analyzing the insufficient coverage of existing benchmarks for long-term change, compound perturbation, and open-world generalization.
- Conducting model fine-tuning and testing around appearance-variation scenarios, with preliminary progress.

Project Experience

Modular Robot System Development Nov 2023–Aug 2025

National 1st Prize / Provincial Project

- For children's programming education, co-designed the architecture of a modular intelligent-hardware system and integrated two-bus power/communication platform, with a main controller scheduling modules and synchronizing power, data, and state over the bus.
- Owned module coordination, bus arbitration, hot-plug discovery, and bus merging, enabling topology rediscovery and stable communication after modules were added, removed, or reordered; also supported protocol/driver integration, three PCB iterations, 24V adaptation, and long-duration tests.
- Measured a 98.5% communication success rate and about 40% higher efficiency than the original design; won the National First Prize (Top 6) at the National University IoT Design Competition and completed provincial acceptance.

A4R: Guided Smartphone AI Assistant May 2025–Aug 2025

National 2nd Prize

- Addressed passive-response failures under ambiguous requests by co-designing A4R (Ask for Refine), changing "execute first" into a clarify-execute-verify loop.
- The Executor proposes actions, Discern filters screen-state evidence, Guide asks questions for missing slots, and Validate checks expected state transitions.
- Built 150+ real interactions across 6 domains and 16 apps; against MobileAgent-V2, MobA, and GUI-Explorer on China Unicom tasks, A4R achieved 64.68% completion, 31.69% skip, and 72.33% effective guidance rates.

Xiaomi Quadruped Robot Intelligent Logistics Delivery System Mar 2025–Aug 2025

Xiaomi Cup National 3rd Prize

- For intelligent logistics delivery in complex environments, built a staged pipeline on the Xiaomi quadruped platform covering departure, pickup, path following, bin recognition, unloading, return, and second delivery.
- Contributed to system design, algorithm debugging, task-flow optimization, and on-site testing; optimized state transitions and action timing to limit the downstream impact of recognition failures and path deviations.
- **Closed-loop validation:** Ran multi-round tests of route stability, bin-recognition robustness, time-limited completion, and recovery after recognition or navigation failures; won the Xiaomi Cup National Third Prize.

A2A Protocol Multi-Agent System Development Nov 8–10, 2025

Shanghai Autumn Camp

- On Holos and A2A v0.3.0, inserted GuideAgent before the default Planning + Routing stage to convert ambiguous requests into field-complete deterministic tasks.
- Chained business-agent execution, ValidateAgent verification, and DiscernAgent extraction; implemented A2A JSON-RPC adaptation, SSE streaming, contextId propagation, task-fingerprint deduplication, LLM/rule fallback, and multi-agent logging.
- **System validation:** Integrated weather, itinerary, travel-demand analysis, and email agents end to end, checking task-state, contextId, and SSE-response consistency across multi-turn collaboration.

SmartPhone Battery: The Moment of Depletion Jan 2026–Feb 2026

MCM Finalist

- For 2026 MCM Problem A, built a unified physical layer mapping screen, CPU/GPU, communication, and sensor loads to total power, with SOC evolution constrained by coulomb-counting and thermal-balance dynamics.
- Paired a 2RC ECM-EKF for real-time SOC estimation and drift correction with a physics-informed neural network using the same dynamics for nonlinear fitting, factor attribution, and cross-validation.
- Validated on 8 heterogeneous sources and 120,000+ time steps: PINN SOC RMSE reached 1.24% and overall TTE MAPE 8.7%; received the 2026 MCM Finalist Award.
- **Comparative analysis:** Used cross-source errors to map prediction stability across load settings and the applicability boundaries of SOC and TTE estimation.

Awards & Honors

1. Representative Competition Achievements

- 2025 National University IoT Design Competition (Huawei Cup), **National First Prize** (Top 6)
- 2025 China Youth Sci-Tech Innovation Challenge, **National Second Prize**
- 2025 National University Computer System Capability Competition, Intelligent System Innovation Design Track (Xiaomi Cup), **National Third Prize**
- 2026 Mathematical Contest in Modeling, **Finalist**
- 2025 Mathematical Contest in Modeling, **Honorable Mention**
- 2025 National University Mathematical Modeling Competition, Hubei Division, **Second Prize**
- 2026 19th National University Software Innovation Competition, South China Regional Contest, **Third Prize**
- 2024 National University IoT Design Competition, Central-Southwest Regional Contest, **First Prize**
- 2024 AIGC Youth Student Employment & Entrepreneurship Promotion Action, Application Development Track (Preliminary), **First Prize**

2. Scholarships & University Honors

- 2025 Wuhan University **Zheng Geru First-Class Scholarship** (RMB 10,000; first in the grade-level defense)
- 2025 Wuhan University **First-Class Scholarship** (top three in the major by comprehensive evaluation)
- 2024 Wuhan University **First-Class Scholarship** (top three in the major by comprehensive evaluation)
- 2025 Wuhan University **Merit Student**
- 2024 Wuhan University **Merit Student**
- 2026 Wuhan University **Active Participant in Social Activities**
- 2025 Wuhan University **Outstanding Student Cadre**
- 2024 School of Computer Science, Wuhan University **Outstanding Communist Youth League Cadre**
- 2024 School of Computer Science, Wuhan University **Outstanding Student Cadre**
- 2024 School of Computer Science, Wuhan University **Outstanding Communist Youth League Member**
- 2025 Wuhan University **Advanced Individual in Extracurricular Academic and Innovation Competitions**
- 2025 School of Computer Science, Wuhan University **Outstanding Communist Youth League Cadre**
- 2026 School of Computer Science, Wuhan University **Outstanding Communist Youth League Member**

3. Extracurricular Achievements

- 2025 Hubei Provincial Radio Direction Finding Championship, Individual All-Around M21-A, **3rd Place**
- 2025 Hubei Provincial Radio Direction Finding Championship, 3.5MHz M21-A, **2nd Place**
- 2025 Hubei Provincial Radio Direction Finding Championship, 144MHz M21-A, **4th Place**
- 2025 Wuhan University Sunshine Sports **Outstanding Individual**

Student Activities

- **Vice Minister, Publicity Department, Student Party Branch Secretaries Joint Council, Wuhan University (Sep 2024–Sep 2025):** Participated in undergraduate Party-building publicity work, assisted in article organization, content coordination, and publishing process optimization; contributed to over 100 published articles including theoretical pieces, promoting standardized publicity operations.
- **Project Team Member, Second Classroom Center, Wuhan University Youth League Committee (Oct 2023–Sep 2024):** Participated in platform quality monitoring and issue investigation, assisted in reporting and following up on multiple platform issues, supporting platform stability optimization and user experience improvement.

Core Skills

- **Programming Languages & Tools:** Python, C/C++, Java; Git, MATLAB, and experiment-management workflows
- **Computer Vision & 3D Modeling:** ReID, SMPL, and 3D Gaussian Splatting; experimental design, ablations, evaluation, and academic writing
- **LLMs & Multi-Agent Systems:** System design, orchestration/routing, verification, and observability; GPT, Claude, and Gemini APIs
- **Embedded Systems & Research:** Embedded drivers, modular protocols, RISC-V/ESP32 integration; literature review, experimentation, and paper writing
- **English:** CET-6 506, CET-4 595; academic reading and technical writing